

吕强



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博士研究生（直博）



中国科学院大学



北京

教育背景

2027.06	中国科学院大学·计算机科学与技术学院
2020.09	计算机应用技术, GPA: 3.76/4.0 • 博士研究生（直博）
2020.06	武汉大学·计算机科学与技术学院
2016.09	计算机科学与技术, GPA: 3.71/4.0 • 工学学士

实习经历

- 阿里云（研究实习生；2025.11–2026.06）：提出 StyleCom，利用精确配对监督与 60 万图像数据集实现基于训练的跨域图像合成，相关一作论文录用于 ACM MM 2026；探索基于 Causal Forcing 的少步视频蒸馏，并通过多模态图像理解与 Prompt Engineering 优化 Qwen-Image 系列模型的图像生成效果。
- 阿里优酷（研究实习生；2024.11–2025.11）：提出 Video-Aligner，由原始视频分支提供教师信号，针对时空退化视频分支设计答案与推理一致性奖励；Video-Holmes 准确率 38.3%，高于主要基线 Video-R1 (36.5%)。提出 Multi-GRPO，以早期树状采样提升生成质量；构建 OCR-Color-10 颜色约束文本渲染数据集，用于多奖励文生图强化学习，并提出分奖励优势估计方法。
- 右脑科技（算法实习生；2024.10–2024.11）：基于 Stable Video Diffusion，以口部遮罩引导语音驱动的唇形同步视频生成。
- 旷视科技（算法实习生；2024.07–2024.09）：基于 Open-Sora，以 BEV 场景布局作为生成条件，探索道路结构可控、时序连贯的驾驶视频生成。

研究方向

- 少样本学习：基于组合原型与可迁移表征开展少样本分类研究，成果发表于 AAAI 2023。
- 可控图像与视频生成：研究跨域图像合成、BEV 条件视频生成、语音驱动唇形同步及少步视频蒸馏。
- 多模态模型强化学习：研究 GRPO 在视频推理与文生图模型后训练中的应用。

专业技能

编程与工具：熟练使用 Python、PyTorch、Hugging Face、Git、LaTeX。

AI 辅助编程：Claude Code、Codex 等 AI 编程工具使用经验丰富。

已录用论文

- Qiang Lyu, Tao Yang, Kunze Wu, et al. “StyleCom: Taming Style Transfer and Collage Prior for Training-Based Cross-Domain Image Composition.” ACM MM 2026 (CCF A 类)。
- Qiang Lyu and Weiqiang Wang. “Compositional Prototypical Networks for Few-Shot Classification.” AAAI 2023 (CCF A 类)。
- Yichen Liu, Donghao Zhou, Jie Wang, Xin Gao, Guisheng Liu, Jiatong Li, Quanwei Zhang, Qiang Lyu, et al. “HiFi-Inpaint: Towards High-Fidelity Reference-Based Inpainting for Generating Detail-Preserving Human-Product Images.” CVPR 2026 (CCF A 类)。
- Junyuan Xiao, Dingkan Liang, Xin Zhou, Yixuan Ye, Tongtong Su, Guangmo Yi, Bin Xia, Qiang Lyu, et al. “Divide and Conquer: Decoupled Representation Alignment for Multimodal World Models.” ECCV 2026 (CCF B 类, 计算机视觉顶级会议)。

在投论文

- Qiang Lyu, Zicong Chen, Chongxiao Wang, et al. “Video-Aligner: Reinforcing Video Reasoning with Consistency Rewards from Information Alignment.” 在投 IEEE TCSVT。
- Qiang Lyu, Zicong Chen, Chongxiao Wang, et al. “Multi-GRPO: Multi-Group Advantage Estimation for Text-to-Image Generation with Tree-Based Trajectories and Multiple Rewards.” 在投 Pattern Recognition; arXiv:2512.00743。

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Ph.D. Candidate



University of Chinese Academy of Sciences



Beijing

Education

Jun. 2027	School of Computer Science and Technology, University of Chinese Academy of Sciences
Sep. 2020	Ph.D. Candidate (direct-entry program) in Computer Application Technology, GPA: 3.76/4.0
Jun. 2020	School of Computer Science and Technology, Wuhan University
Sep. 2016	Bachelor of Engineering in Computer Science and Technology, GPA: 3.71/4.0

Internship Experience

- **Alibaba Cloud** (Research Intern; Nov. 2025–Jun. 2026): Proposed StyleCom, a training-based cross-domain image composition framework with precisely paired supervision and a 600k-image dataset; first-author paper accepted at ACM MM 2026. Explored few-step video distillation with Causal Forcing and combined multimodal image understanding with prompt engineering to improve image generation quality in the Qwen-Image model family.
- **Alibaba Youku** (Research Intern; Nov. 2024–Nov. 2025): Proposed Video-Aligner, using outputs from the original-video branch as teacher signals to design answer and reasoning consistency rewards for the temporally and spatially degraded video branch; achieved 38.3% on Video-Holmes versus 36.5% for the main baseline, Video-R1. Proposed Multi-GRPO, using early tree-based trajectories to improve generation quality; curated OCR-Color-10, a color-constrained text-rendering dataset for studying multi-reward text-to-image RL, and introduced per-reward advantage estimation to balance objectives.
- **RightBrainAI** (Algorithm Intern; Oct.–Nov. 2024): Explored speech-driven lip synchronization based on Stable Video Diffusion, using mouth-region masks to guide video generation.
- **Megvii** (Algorithm Intern; Jul.–Sep. 2024): Explored driving video generation based on Open-Sora, using BEV scene layouts as conditioning signals to achieve controllable road structures and temporal coherence.

Research Areas

- **Few-shot learning**: compositional prototypes and transferable representations for few-shot classification (AAAI 2023).
- **Controllable generation**: cross-domain image composition, BEV-conditioned video, speech-driven lip synchronization, and video distillation.
- **Multimodal reinforcement learning**: applications of GRPO to video reasoning and text-to-image model post-training.



Skills

Programming and tools: Proficient in Python, PyTorch, Hugging Face, Git, and LaTeX.

AI-assisted coding: Extensive hands-on experience with Claude Code and Codex.

Publications

- **Qiang Lyu**, Tao Yang, Kunze Wu, et al. “StyleCom: Taming Style Transfer and Collage Prior for Training-Based Cross-Domain Image Composition.” *ACM MM 2026* (CCF A).
- **Qiang Lyu** and Weiqiang Wang. “Compositional Prototypical Networks for Few-Shot Classification.” *AAAI 2023* (CCF A).
- Yichen Liu, Donghao Zhou, Jie Wang, Xin Gao, Guisheng Liu, Jiatong Li, Quanwei Zhang, **Qiang Lyu**, et al. “HiFi-Inpaint: Towards High-Fidelity Reference-Based Inpainting for Generating Detail-Preserving Human-Product Images.” *CVPR 2026* (CCF A).
- Junyuan Xiao, Dingkan Liang, Xin Zhou, Yixuan Ye, Tongtong Su, Guangmo Yi, Bin Xia, **Qiang Lyu**, et al. “Divide and Conquer: Decoupled Representation Alignment for Multimodal World Models.” *ECCV 2026* (CCF B; premier computer vision conference).

Submitted Manuscripts

- **Qiang Lyu**, Zicong Chen, Chongxiao Wang, et al. “Video-Aligner: Reinforcing Video Reasoning with Consistency Rewards from Information Alignment.” Submitted to *IEEE TCSVT*.
- **Qiang Lyu**, Zicong Chen, Chongxiao Wang, et al. “Multi-GRPO: Multi-Group Advantage Estimation for Text-to-Image Generation with Tree-Based Trajectories and Multiple Rewards.” Submitted to *Pattern Recognition*; arXiv preprint 2512.00743.